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A Forced-Oscillation Source Locating method based on Dissipating Energy Flow in Generator-Connected Lines

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Abstract

Vietnam needs to continue integrating renewable energy sources into the power grid to progressively replace fossil-fueled generation in order to achieve emission-reduction targets in the near future. The growing penetration of renewables, along with the involvement of excitation systems, governors and power system stabilizers (PSS), increases the risk of forced oscillations (FO) and raises the need to locate those sources on the grid. Improper tuning or operation of these devices, as well as low damping ratios in systems with high renewable penetration, can adversely affect power system stability. In this paper, the Dissipating Energy Flow method is introduced to identify the source of forced oscillations, which are often generated on generators, loads or FACTS devices. The method operates on phasor measurement unit (PMU) sample data and is assessed on the two-area Kundur test system. Results show that this method achieves high accuracy for different forced oscillation frequency scenarios, even when the forced oscillation frequency is close to the natural oscillation mode. Therefore, this method can serve as a basis for early detection and mitigation of oscillation sources.

Keywords: Dissipating Energy Flow, forced oscillation, oscillation source location.

1. Introduction

Nowadays, with the increase of inverter-based resources in electricity generators such as wind and solar energy, the total inertia of the power system will also decrease. Therefore, the capability to deal with grid instabilities will also decrease due to the lack of damping torque ratio. The urgency to identify sources of oscillations for appropriate adjusting measures is also increasing. If the culprit causing oscillations on the grid cannot be eliminated quickly, the grid consequently become sensitive to disturbances and might reduce reliability. Currently, on typical grids, there are two common types of sustained oscillations, which are Forced Oscillations (FO) and Natural Oscillations.

In the context of increasingly large-scale power systems and the penetration of Inverter-Based Resources (IBRs), original model-based grid analysis approaches are more difficult, requiring detailed models and continuous updates. In addition, the operating point changes over time, requiring repeated linearization and parameterization, along with the effects of limits or nonlinearities, which reduce the reliability of small signal analysis. In contrast, methods aimed at online determination based on PMU measurements exploit the current operating signal directly, are less dependent on the model parameters, thereby allowing early source-area identification or detection of the excitation source even when the grid configuration and the transmission power change rapidly.

There have been many approaches in the world to the problem of determining this oscillation source such as methods based on traveling wave [1], [2], damping torque [3], mode shape estimation [4], [5] or energy-based [6]. These methods all have their own advantages and drawbacks. Some methods always determine correctly but do not always give results. Some other approaches can give many results, but still include correct results. Some methods cannot even find the oscillation source or give wrong conclusions. For example, the traveling wave method often depends on choosing a proper time window and requires a real-time wave map. These are often not available and often inaccurate due to the change of parameters of components in power system. Not only that, the damping torque method requires detailed generator parameters, which makes the assumptions simplifying the generator model incorrect and can cause errors in the calculation of forced oscillation flow direction. Moreover, the mode shape estimation method can indicate the regions with high accuracy of containing the oscillation source. However, when the FO frequency is close to the inter-area mode, this method can be disturbed and therefore inaccurate. Therefore, according to [7], the energy-based method is the method that has been interested and researched recently.

The article is divided into 5 parts. Part 1 summarizes some information about the current power system problem and the necessity of detecting forced oscillations on the grid as well. Part 2 presents the

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theoretical basis of the dissipating energy flow and some simulation theories. The model layout set up for running simulations will be introduced in part 3. Finally, in parts 4 and 5, the simulation results will be clarified in scenarios, along with comparisons to give some conclusions at the end.

2. Methodology and theorectical basis

2.1. Process

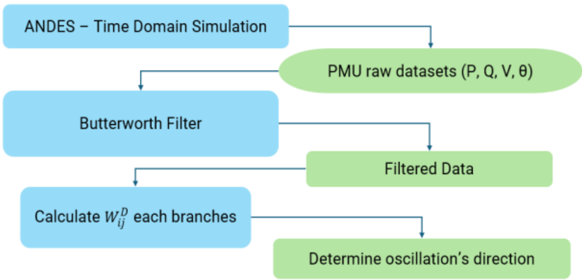

Figure 1. Problem implementation process

The implementation process of the problem is described as shown in the figure above. First, time domain simulation is done to collect grid data such as amplitude and phase angle of voltage, current, active power P, reactive power Q from PMUs at buses and branches. After having the data, the authors clean the data by using Butterworth band-pass filter in order to exclude other oscillation components, isolate the exact oscillation needed for research. The forced oscillation frequency on the grid is assumed to be known because finding the forced oscillation frequency is not within the scope of this research. Next, a dataset in a particular time period is utilized to avoid using all the data, including parameters in the transient period, causing inaccuracies in the calculation. After having a complete dataset, the energy dissipated over time on all branches is calculated based on the traditional energy equation. Finally, the data in each branch is linearized to calculate the slope DEF of the equation and determine the direction of oscillation propagation based on the sign of DEF. From there we can detect the source of oscillation.

2.2. Theory of dissipative energy flow

According to [7] the energy dissipation on branch ij is calculated as follows:

$$W_{ij}^D = \int (\Delta P_{ij} d\Delta\theta_{ij} + \Delta Q_{ij} d\Delta(lnV_i))$$

$$= \int (2\pi \Delta P_{ij} \Delta f_i dt + \Delta Q_{ij} d\Delta(lnV_i)) \quad (1)$$

Where ΔP_{ij} and ΔQ_{ij} are the deviations of the P and Q values from the values on branches ij in the normal state of the system. Δf_i and $\Delta\theta_{ij}$ are the frequency and voltage angle deviations at bus i. V_i is the voltage amplitude at bus i. $\Delta(lnV_i)$ indicates the proportional relation between reactive energy and the variation of voltage.

However, we cannot use this formula directly because the component lnV_i can only be determined with positive values, while the voltage signal filtered through the filter contains both negative and positive values. Thus, the component $dlnV_i$ is approximately transformed to $\frac{d\Delta V_i}{V_i^*}$ with the value $V_i^* = \tilde{V}_i + \Delta V_i$, $\tilde{V}_i$ is the average voltage in the considered interval. This equation is only accurate in the case of only one oscillation mode (single mode). From there, the equation W_{ij}^D becomes:

$$W_{ij}^D \approx \int (\Delta P_{ij} d\Delta\theta_{ij} + \Delta Q_{ij} \frac{d\Delta V_i}{V_i^*})$$

$$\approx \int \left(2\pi \Delta P_{ij} \Delta f_i dt + \Delta Q_{ij} \frac{d\Delta V_i}{V_i^*}\right) \quad (2)$$

When applying the formula (2) to real PMU data, which are discrete signals, Riemann-Stieltjes difference is applied for equation (2):

$$W_{ij}^D = \sum_{k=1}^{N-1} [(\Delta P_{ij}(k)(\Delta\theta_i(k) - \Delta\theta_i(k-1))$$

$$+ \Delta Q_{ij}(k)(\frac{\Delta V_i(k) - \Delta V_i(k-1)}{V_i^*})] \quad (3)$$

The quantities Δ are all post-filtered signals. After having a series of W_{ij}^D values over time, we linearize the $W_{ij}(t)$ values to create a first-degree polynominal equation:

$$W_{ij}^D(t) = DEF_{ij} \times t + b_{ij} \quad (4)$$

From there, we can determine the DEF value, which is the slope rate of the equation after linearizing $W_{ij}(t)$. If this DEF_{ij} value is positive, we can conclude that the oscillation goes from bus i to bus j. Conversely, when DEF_{ij} is negative, the oscillation tends to flow from bus j to bus i.

2.3. Butterworth band-pass

To filter the interested forced oscillation frequency from the original signal, this paper used a Butterworth band-pass filter developed from an analog low-pass prototype. This filter is introduced in [7] as a filtering method accurate enough to be used for the purpose of determining the oscillation source. With the filter function of the low-pass sample:

$$|H_{LP}^{(a)}(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_C}\right)^{2n}} \quad (5)$$

Where $|H_{LP}^{(a)}(j\Omega)|^2$ is the squared amplitude when considered on the frequency axis s = $j\Omega$. Value Ω and Ω_C are the angular frequency (rad/s) and the fixed cutoff frequency (3 dB), n is the order of the filter.

The standard band-pass transform gives us the analog band-pass filter:

$$H_{BP}^{(a)}(s) = H_{LP}^{(a)}\left(\frac{s^2 + \Omega_0^2}{Bs}\right) \quad (6)$$

with $B = \Omega_2 - \Omega_1$, $\Omega_0 = \sqrt{\Omega_1\Omega_2}$

Here, B illustrates the bandwidth, Ω_1 and Ω_2 are the desired passband amplitude (rad/s) of the analog filter. Ω_0 is the geometric center frequency of the passband (rad/s).

However, since the signal obtained through TDS simulation on ANDES as well as the signal received from the PMU is a discrete signal, it is necessary to discretize the filter and then implement it using bilinear transform with sampling period $T = 1/f_s$:

$$s = \frac{2}{T} \times \frac{1 - z^{-1}}{1 + z^{-1}}$$

$$=> \; H_{BP}(z) = H_{BP}^{(a)}\Big|_{s=\frac{2}{T}\times\frac{1-z^{-1}}{1+z^{-1}}} \quad (7)$$

A vital step before filtering is phase unwrapping. The phase angle received from voltage V, which is wrapped up in a period of 2π, leading to spurious jumps at the $\pm\pi$ points and strong noise. This action helps to preserve the amplitude-phase coherence around f_0, stabilizes the direction sign of the $\mathrm{DEF_{ij}}$ index, and reduces the integral noise at the window boundary.

2.4. Time domain simulation

The paper performs Time Domain Simulation running on the ANDES library. This is an open source library in Python programming language for simulating power systems. ANDES supports the calculation of power flow, transient stability or small-signal stability.

TDS is a transient stability simulation for power systems. TDS simulation solves the dynamics of power systems, which are modeled as Differential Algebraic Equations (DAEs). ANDES employs the Implicit Trapezoidal Method (ITM) to solve the DAE equation. This is a well-known second-order numerical integration technique, recognized for its stability and efficiency in handling complex power system models.

The mathematical basis of TDS in ANDES is defined in a semi-explicit form as follows:

$$\begin{cases} T\dot{x} = f(x,y) \\ 0 = g(x,y) \end{cases} \quad (8)$$

Where T is the coefficient matrix, $\dot{x}$ represents the rate of change of the state variables. $f(x,y)$ and $g(x,y)$ depict differential equations depending on the state variables x and the algebraic variables y. Based on these equations, four sparse Jacobian matrices (f_x, f_y, g_x, g_y) are constructed to form the complete Jacobian matrix A_C for the execution of the ITM.

3. Building simulation model

The paper conducts theoretical verification of the energy dissipation flow by simulation and calculation using the open source library ANDES [8] with the Python programming language. A Kundur two-area test system was utilized to verify the accuracy of the method. The generators of the model are equipped with governors and excitation.

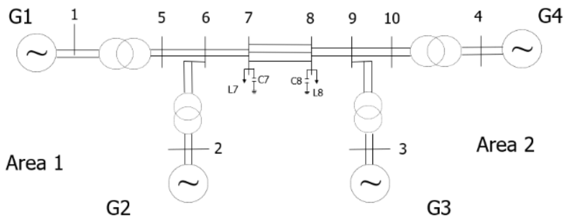

Figure 2. Kundur's test grid structure

The model parameters are described in [9]. The TDS simulation in the time domain was performed for 200 seconds with oscillation injection after 2 seconds. The data taken for the study is the data from the 50s to 150s to eliminate the free response in the transient following the excitation.

This paper chose the fifth order in Butterworth filter. After filtering, the values in equation (3) are initialized, from which the model calculates the values $W_{ij}(t)$.

The time domain simulation results are also fed into the rolling window with mean subtraction method to compare and verify the effectiveness of the proposed dissipating energy flow method.

4. Results and Discussions

The paper simulated two different scenarios in the table 1 below. After running TDS on ANDES, the data was taken with a resolution of 0.01 seconds.

Table 1. Simulation scenarios

Scenario	FO Injecting location	Frequency (Hz)	Amplitude (p.u)
1	Exciter of Gen 3	2	0.02
2	Exciter of Gen 1	0.6	0.02

The signals of the PMU before filtering on branches 7-8 with scenario 1 are as displayed in Figure 3a. After using the Butterworth filter as introduced above, with the P signal at branch 7-8, we get the filtered signal as displayed in figure 3. The authors choose the filter window between second 50 and 150 to ensure that the system had passed the transient phase after disturbance.

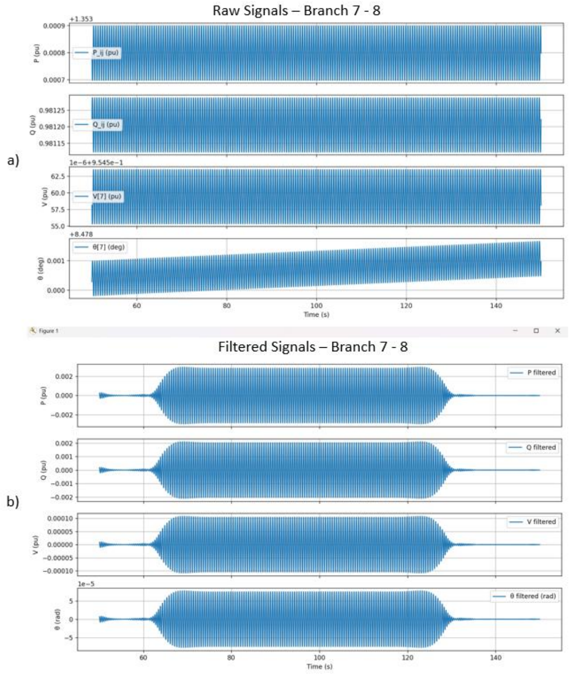

Figure 3. Branch signal 7-8 according to scenario 1

a) Raw signals; b) Filtered

After receiving the filtered signal at the branches through the Butterworth filter, this paper proceeds to calculate W_{ij}^{D} based on formula (3). The figure 4 below shows the trend of W_{ij}^{D} over time in 2 introduced scenarios at 4 branches connected to the generator and the tie-line 6-7 as well.

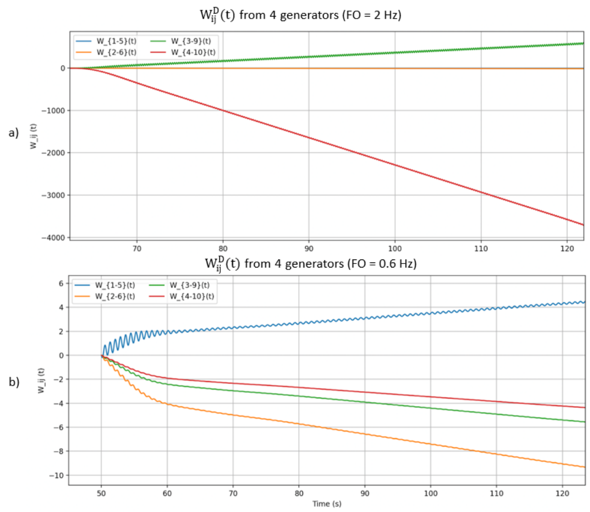

Figure 4. $W_{ij}^{D}(t)$ from 4 generators

a) Scenario 1; b) Scenario 2

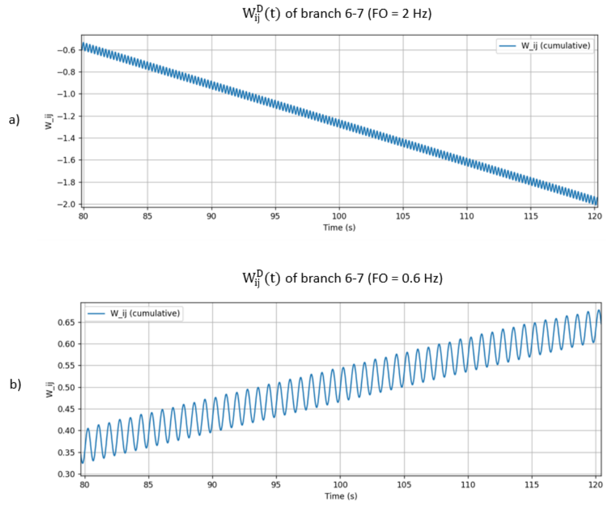

Figure 5. $W_{ij}^{D}(t)$ of branch 6-7

a) Scenario 1; b) Scenario 2

The graphs of W_{ij}^{D} to detect the oscillation direction of all branches are generated. Then this paper proceeded to linearize the values $W_{ij}^{D}(t)$ to calculate the overall trend for the entire line. From there, the DEF value on some typical branches of the 2 scenarios is shown in the following table.

Table 2. Calculated DEF values

Line	DEF in Scenario 1	DEF in Scenario 2
Gen 1 to Bus 5	- 0.0311	**0.0447**
Gen 2 to Bus 6	- 0.1826	- 0.0937
Gen 3 to Bus 9	**7.86**	- 0.0559
Gen 4 to Bus 10	- 51.77	- 0.0437

In addition to the main results presented, to verify the accuracy of the proposed method, the paper compares the results with the alternative method of subtracting the average value on a sliding window and the $W_{ij}^{D}(t)$ in 4 generator-connected lines are illustrated below in Figure 6.

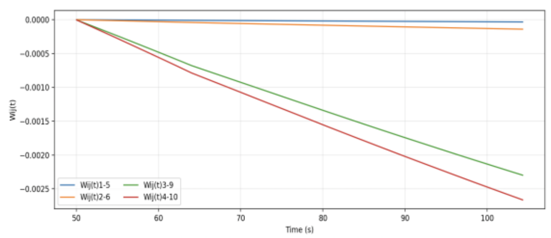

Figure 6. $W_{ij}^{D}(t)$ from 4 generators in Scenario 1.

It is clearly seen that in the 2Hz forced oscillation excitation scenario, all four generators give the same trend of $W_{ij}^{D}(t)$ which decreases over time, which means that all four generators are receiving energy. This is not accurate and is different compared to scenario 1 where generator 3 is the oscillation source. This might cause by the mix of both the natural oscillation mode in the grid and the forced oscillation. Therefore, the result is distorted, leading to failure to detect the generator causing oscillation in the grid.

As a matter of fact, the authors found that the method assists to determine the direction of FO's energy flow on the branches accurately. In all scenarios, the generator causing the oscillation is found correctly and the generators receiving the oscillation are not mistakenly detected.

5. Conclusion

This paper have presented a framework to identify the source of forced oscillation in power system. The method is verified through simulation scenarios on the open source library ANDES with Python programming language. The scenarios that cause forced oscillation are implemented when forced oscillation occurs on the generator's excitation system. The results show that the method has high accuracy and can be the basis for operators and researchers at the first stage of oscillation source identification. This paper highlights the role of the dissipating energy flow in generators method as the first monitoring layer to localize forced oscillation, while grid analysis methods are used in the post-check step to verify the mechanism and propose solutions to eliminate FOs. In addition, this method can be combined with other identification methods to increase the accuracy of the model.

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